

Birla Institute of Technology & Science, Pilani
Work-Integrated Learning Programmes Division
Second Semester 2025-2026

Mid-Semester Test
(EC-2 Regular)

Course No. : SE ZG519
Course Title : Data Structures and Algorithms Design
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : Saturday, 14/03/2026 (FN)

No. of Pages	= 4
No. of Questions	= 4

Note:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1 [Marks: 2 + 3 + 10 = 15]. Consider hashing with double hashing as collision resolution mechanism. Assume that both primary ($h_1()$) and secondary ($h_2()$) hash functions follow a linear equation of the form $y = mx + c$ and the array A has a maximum size of N elements.

a) Express $h_1()$ and $h_2()$ appropriately to calculate the hash index for an input key x .

b) Express the formula to calculate the hash index value in the array A for a given key x that uses double hashing mechanism. To write your answer, fill in the blank in the following:

$A[\quad] = x$

c) Consider that you need to insert the following sequence of values into a hash table, where $N = 12$, $h_1()$ has $m_1 = 3$ and $c_1 = 3$, and $h_2()$ has $m_2 = 2$ and $c_2 = 5$, using double hashing mechanism.

Input: 3, 2, 9, 6, 11, 13, 7, 12, 3, 6

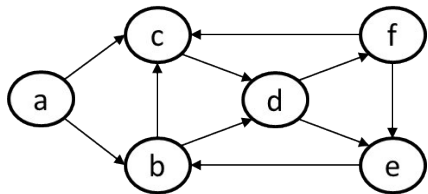
To answer this question, fill in the following table: (first row is filled as a sample)

Key	$h_1(k)$	$h_2(k)$	#probe (j)	Final Index
3	0		1	0
2				
9				
6				
11				
13				
7				
12				
3				
6				

Q2 [Marks: 3 + 12 = 15]. Consider traversing a graph using breadth first search (BFS).

a) Write the algorithm for BFS in plain English (**no pseudocode please à pseudocode will not fetch any mark**). Use the standard functions as discussed in the lecture classes for any data structure that you may need to construct the algorithm of “**Algorithm BFS(startVertex)**”.

b) Consider the following graph G.

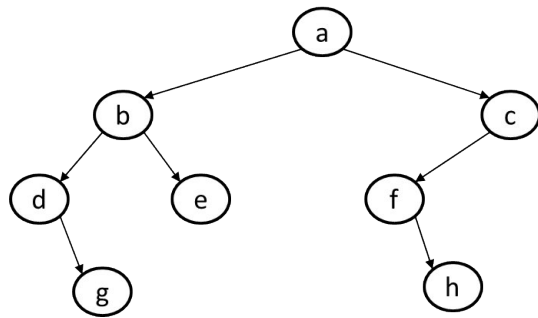


Run your BFS algorithm on G and show the output of your BFS using the following table. The initial step is shown to start your execution. **Note: Whenever you have multiple adjacent nodes, insert them in alphabetic order, e.g., if v1 and v2 are adjacent nodes, then insert them as v1 followed by v2 if v1 precedes v2 in the alphabetic order.**

Visited						
Queue	a					

Every time a node is visited, show the status of Visited and Queue by drawing them separately. Hence, you need to show six drawings, i.e., tables.

Q3 [Marks: 3.5 + 3 + 8.5 = 15]. Consider the following binary tree.



a) Show array representation of this binary tree. Use the following table to show your answer, one entry is filled for you to start your answer.

[illegible]

b) Fill in the blanks in the following sketch of an algorithm that solves the problem of searching a key using recursion. **Certain hints are provided as comments to fill in the blanks.** **Note:** you need to fill in the blanks indicated by *B1*:, *B2*:, .. and so on. Write only those statements that can fill in the blanks, hence you need to write a total of six statements, i.e., pseudocodes, at least.

Search (root, key): Returns index of Key in tree

```
//Array A stores the tree
```

$$i \leftarrow \text{root}$$

```
if (Bl: ) then // Key found
```

```
return(B2:      ) // Returns the index
```

else

```
if ( $B3:$            ) then // Condition to Traverse right subtree
```

```
Search(B4:      ) // Traverse right subtree
```

```

if ( $B5:$   $\text{root} \neq \text{null}$  ) then //condition to traverse left subtree

```

```
Search (B6:      ) // Traverse left subtree
```

```
return(-1)           // Key not found, i.e., invalid index
```

c) Use the algorithm in part b) to search for key = g in the tree. You need to show the intermediate output of the algorithm in terms of the index of root node and its value as you perform the recursive calls using the following table. First entry in the table is filled to start your answer. **Leave blank in the row “Node” if there is no value for a root index, marks for leaving blank spaces will be awarded only if all the blank spaces are correct.**

[illegible]

Q4 [Marks: 6 + 9 = 15]. Consider the evaluation of arithmetic expression using infix to postfix conversion.

a) Fill in the blanks (indicated by **B1:**, **B2:**, and so on) in the following algorithm snippet for converting an infix expression to postfix expression. You may use hints provided as comments.

```

Algorithm InfixToPostfix(input, len):
    input[len + 1]  $\leftarrow$  ')'           // insert ')' as the last element in the array
    B1:                               //
    while (B2:           ) do         // stack s is not empty
        if (B3:           ) then
            print input[i]
        else if input[i] = ')'
            while (B4:           )do
                print x
            else if (Priority(input[i]) <= Priority(TOP(s))) //stack s contains only operators
                while Priority(input[i]) <= Priority(TOP(s)) do
                    B5:
                    print x
                Push(input[i])           //input[i] is an operator
            else
                B6:
            B7:

```

b) Using the above algorithm to convert **$a+((b-c)-d)$** to its postfix expression. To write your answer, fill in the following table (first row is filled to initiate the answer). For each input symbol, update the stack and output columns.

Step#	Input	Stack	Output
1	a+((b-c)-d)		
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			